COLLEGE OF ENGINEERING, UIC

CS 401/MCS 401: Computer Algorithms I

Course Syllabus – Fall 2026 – 3 undergraduate hours – 4 graduate hours

Students are expected to be familiar with all of the information below, and are responsible for knowing all policies detailed in this document. Read it fully and carefully. **This document has multiple tabs. Click the outline button in the top left to view them.**

**This document may change, refer to online copy only. Last updated 8/5/2026.*

I. Instructors & Course Details

Omar Ibrahim

Email address: oibra@uic.edu
Drop-In Office Hours: TBD in CDRLC 3460

Teaching Assistants:

TBD

Canvas Course Site

Course information, documents, resources, blank and inked slides will all be posted on the course Canvas, in addition to lecture recordings and your grades. Students are expected to log into the course site regularly to learn about any developments related to the course. We do not communicate through Canvas – please use Piazza instead. Do not contact the course staff through Canvas; you are not guaranteed a response. For all technical questions about Canvas, email the Learning Technology Solutions team at LTS@uic.edu.

Course Modality and Schedule

This course is taught on campus. This class is crosslisted as CRNs 10669 (CS 401 for undergrads), 20262 (CS 401 for grads), 12374 (MCS 401 for undergrads), and 20263 (MCS 401 for grads). All 4 CRNs are one class, and will all meet for the same lecture. If you are not registered for one of these CRNs, you are in a different section of the course and nothing described in this document is guaranteed to apply to you.

Lecture	TR	2:00-3:15 pm	LC F4

II. Course Information

Catalog Course Description

Design and analysis of computer algorithms. Divide-and-conquer, dynamic programming, greedy method, backtracking. Algorithms for sorting, searching, graph computations, pattern matching, NP-complete problems.

Prerequisite(s):

- Grade of C or better in MCS 306 or CS 251

If you are a graduate student who has not taken one of these classes, you will still be expected to have mastery over material covered in these classes and their prerequisites; in particular, discrete mathematics (particularly functions, set theory, proofs, and induction) and basic data data structure (including recursion and graphs).

Course Goals and Learning Outcomes

CS 401 is an introduction to computer algorithms. By the end of this course, students will be able to:

- Model real world problems as computational problems and using data structure representations
- Determine an appropriate algorithm design paradigm for a new problem
- Design an algorithm using a number of different algorithm-design paradigms (including greedy algorithms, divide-and-conquer, dynamic programing, and network flow)
- Prove that an algorithm produces the correct answer
- Reduce between a new problem and a known problem (both for reusing an existing algorithm and for showing computational hardness)
- Identify and design for problems that are infeasible to compute
- Consider the implications of computational modeling decisions in the real world

This is a required course for students majoring in Computer Science. A brief list of topics to be covered is in the schedule below.

Growth Mindset:

Course materials and assignments can be complex and challenging, but they are crucial to your intellectual and personal growth and development. There are times you may need extra help. Students who attend class consistently, complete all assignments, thoughtfully engage with feedback on work, develop good study strategies, visit the tutoring center, and contact faculty

when they are struggling can develop a thorough understanding of the course material and ultimately succeed in the course!

Piazza

Piazza will be the primary method of communication from and to the course staff. All announcements will be made on Piazza and Canvas.

Sign yourself up using this link:

On Piazza, you may make public posts for questions that can benefit everyone, such as clarification of policies or questions about content and assignments. You may also make private posts to the entire instructional staff or specific instructors or TAs if you have a personal concern.

When posting, you should follow these guidelines:

- Post publicly when possible so other students can benefit from the question. You may post anonymously to other students (but the teaching staff will still see your name).
- Search before asking; your question may have already been answered.
- If you know the answer to someone's question, feel free to answer!
- Don't post answers to assignment questions; instead, offer hints or clarifications.
- All posts should be directly related to CS 401. If your student organization wishes to post an unrelated announcement, please ask the instructors first. Any unauthorized posts will be deleted.
- You can expect a response within 1 business day of your question.

We do not use email communication for this class. Emails sent to the instructor and TAs can easily be missed, delayed, or lost. Since our class is a large system of teachers and students, email does not work, and we ask that you do not use it. Class communications from instructors will be sent as Canvas announcements (with force email turned on).

Gradescope

We are using Gradescope.com for submission and grading of check-ins, homework assignments, and exams.

Office Hours

For any questions related to course material or coursework, your first place to go should be office hours. The course staff is here to help you with the course; please let them. An office hours schedule will be available here and linked on Canvas and Piazza by the second week of classes.

Recommended Textbooks

There are no required textbooks. The following textbooks/reference books are recommended, and are available from the course Canvas page.

- Algorithm Design by Kleinberg and Tardos
- Algorithms by Jeff Erickson

Any relevant readings from these texts will be made available to students. You are not required to buy either textbook.

Technology

If needed, computer labs are available throughout campus, with some spaces open 24/7 (https://teaching.uic.edu/ed-tech/learning-spaces/computer-labs/). Short-term equipment lending, for up to three business days, is available to students as well (https://teaching.uic.edu/ed-tech/equipment-lending/short-term-lending/).

Respect for Copyright

Please protect the copyright integrity of all course materials and content. Please do not upload course materials you did not create to third-party websites, including GenAI tools, or share content with anyone not enrolled in our course.

III. COURSE POLICIES & CLASSROOM EXPECTATIONS

Grading Policy and Point Breakdown

All graded work in the course is assigned a certain number of points. Your final course grade will be determined based on the total number of points you have received:

Total Points	Minimum Guaranteed Grade
360	**A**
320	**B**
280	**C**
200	**D**

Note that this only tells you the minimum grade that you are guaranteed to receive in the course. In the event that coursework turns out more difficult than intended, we may make the grade cutoffs more generous than indicated here, but we will not make them less generous. Note that there is no guarantee that the cutoffs for each grade will be any lower than presented here.

Course Work

Lecture Check-Ins

Your attendance and participation in lectures does not contribute to your course grade (*although you are still strongly encouraged to attend, and considered responsible for all material and announcements from lectures*). However, this course moves quickly and we're not always able to get quick feedback on how students are doing with the course material. There will be a "Check-In" assignment released at the beginning of each week due by Friday of that week. These are meant to be a pulse check for your understanding of that week's course material and are designed to be completed within ~20-30 minutes. You should complete the Check-In before starting a homework assignment released that week.

There will be ~14 total Lecture Check-Ins this semester (there will be no Check-In week 14). Each Check-In is worth **3 points**.

Exams

We will have two exams; a midterm exam and a final exam. The midterm exam will be given in class, tentatively on **10/08/2026** (subject to change). The final exam date and time are usually announced around week 10. Each exam will be worth **100 points**.

We do not offer any make-up if you miss an exam. We *will* offer an alternate exam to students who need to miss the scheduled exam for documented and legitimate conflicts that we are notified of **in advance**. If you need to take the alternate exam, please send a private Piazza message as soon as you are aware of the conflict. If you experience an emergency on the day of the exam, please contact us ASAP; we will also ask for documentation and offer you the opportunity to take the same alternate exam.

Homework Assignments

There will be 7 graded homework assignments distributed approximately bi-weekly throughout the semester. There are three types of problems on homework assignments: mechanical, long-form, and programming.

- **Mechanical problems** will usually ask you to execute an algorithm or give short answers about an algorithm (like an input that causes bad behavior). These problems are usually shorter and require a surface-level understanding of what was covered in lecture. There will be 1 of these problems per homework.
- **Long-form problems** generally involve designing an algorithm – usually by hand, in pseudocode. Long-form problems also might ask you to consider real-world effects of using algorithms or to prove facts about algorithms. These problems usually require applying the contents of lectures to new scenarios. There will be approximately 2-3 of these problems per homework.
- **Programming problem**s generally involve implementing an algorithm in real (Python) code. These questions will be autograded on gradescope, and some

(non-comprehensive) tests for them may be made available for you there as well. There will be approximately 0-1 of these problems per homework.

Mechanical and long-form problems will be graded by the entire course staff. Programming problems will be autograded by Gradescope. Each homework assignment will be worth **24 points**.

Please see the **Homework Policies** tab for advice and details about the logistics, format, form, and content expected for submitting homework assignments.

Homework Collaboration

The collaboration policy differs for graduate and undergraduate students. Please see the relevant section for you below.

Graduate Students

Graduate students are required to submit individual solutions to the homework. You may still discuss general ideas and concepts with other students, but your exact execution must be your own, and your submitted solution must be written up entirely individually.

Undergraduate Students

Groups of **up to three** undergraduate students are allowed to submit a single solution for each homework assignment. Every member of the group receives the same grade and the same credit for the entire assignment. That means every member of the group is responsible for the entire assignment.

Group solutions must represent an honest collaborative effort by all members of the group.

In particular, groups must *not* delegate one problem to each group member; only the students who actively worked on a problem may add their names to the solution. **At a minimum, you must *read, understand, and approve* anything submitted with your name on it. Allowing someone else to add your name to a solution to which you made no contribution is plagiarism.** This does not mean that every student in a group must contribute *good ideas* or must help in the actual *writing* of every group solution. Asking "stupid" questions, proposing bad ideas, shooting down bad ideas, working out examples (even if they don't appear in the solution), uncovering bugs, and even just acting as a sounding board for other group members are all legitimate contributions.

If a submitted homework contains plagiarized material, we will separately determine each student's culpability (if any) and penalty (if any), in accordance with the Student Code. **By default, every member of the homework group will be given the same penalty.** (Again, this

is the same standard that is applied to coauthors of research papers.) If you cheat, you are not only endangering your grade, and possibly your academic career, but your colleagues' as well.

Late Homework

Homework Assignments will typically be released on a Tuesday and **due Thursday of the following week at 11:59pm**. Each student will receive **5 late days** that they can use on homework submitted through Gradescope. Each late day allows you to submit an assignment up to 24 hours later than it would normally be due. **You may use at most 1 late day on any assignment**. There is no benefit to retaining late days at the end of the semester and late days are deducted automatically. Late days will not be used if a student does not submit an assignment. **We will not accept homework more than 24 hours late for any reason.**

Late days are intended to handle the "normal" difficulties during a semester (midterms in another class, family birthday party to attend, bad colds). If you have a more extreme situation (e.g., an extended illness or a family emergency) or run out of late days, contact Omar (via email or Piazza post) for accommodations.

Other Course Policies

Academic Integrity

As a student and member of the UIC community, you are expected to adhere to the Community Standards of academic integrity, accountability, and respect. Please review the UIC Student Disciplinary Policy for additional information.

We do not recommend that students use Gen AI tools to complete homework assignments, but it is not forbidden. Students must understand and be able to explain all solutions that they submit for homework assignments. The homework assignments are of low weight for the overall course grade, and are fundamentally tools for learning the course material. The exams will be written to measure whether students understand the course material, and may include questions designed to directly evaluate whether students understand previously completed homework assignments.

Remember that your goal as a student in this class is to learn to solve problems, not to collect points. The only way to learn to do the thing is to actually do the thing; watching someone else do the thing isn't enough.

If you really need points for motivation, remember that points on the exams are worth significantly more than points on the homework. Regardless of whether it constitutes plagiarism, or whether you get caught, getting too much help on your homework will hurt your final grade. If you don't learn how to solve algorithmic problems on your own, you will perform poorly on the exams, which make up a significant portion of your grade.

For more thorough expectations and details, please see the Academic Integrity tab.

Penalties

The default penalty for cheating on a homework problem will be a 0 on that entire homework assignment. The default penalty for cheating on an exam will be a 0 on that exam and a referral to the Office of the Dean of Students. The penalty for a second offense of any kind will be a referral to the Office of the Dean of Students. Note that multiple referrals to the Office of the Dean of Students, even from different classes, can result in probation or suspension.

Email Expectations

Students are responsible for all information instructors send to their UIC email and Canvas accounts. Faculty messages should be regularly monitored and read in a timely fashion. We assume you check your email regularly, at the very least once every 24 hours. We will post all class announcements on Piazza and on Canvas announcements (we will force send an email from Canvas announcements).

Student Behavior

Since a portion of the class is discussion-based, it is vital that each of you treats the others in the class with respect. Of course, we love to hear your opinions, but make sure that others have the chance to speak as well. Be kind to one another and respectful of others' opinions, even if they disagree with your own.

Our class will follow the **CS Code of Conduct**. If you are not adhering to our course norms, a case of behavior misconduct will be submitted to the Dean of Students and to the Director of Undergraduate Studies in the department of Computer Science. If you are not adhering to our course norms, you will not get full credit for your work in this class. For extreme cases of violating the course norms, credit for the course will not be given.

Regrade Requests

Grading mistakes do happen, so it is in your best interest to review all returned materials and published scores for correctness. When those mistakes happen, we want to correct them. Regrade requests for homework assignments and exams will be submitted through Gradescope. To request a regrade, you must follow these guidelines:

- Regrade requests should address the TA feedback and your solution, e.g., "The feedback says that x was not introduced as arbitrary, but x is introduced as arbitrary on the second line of the proof."
- Regrade requests must be specific---regrades simply requesting more partial credit without reference to the specific rubric/feedback and submission will be rejected.
- Please remember that people are reading your regrade requests, and phrase regrade requests professionally and respectfully.

If you do not follow these guidelines, the teaching staff will dismiss your regrade request without further review. Regrade requests that are rude or disrespectful will also be dismissed

without further review. *We do not change the rubric by which your work is graded by for a regrade request.* TAs will respond to regrade requests when they can - note that this is not always quick.

Once a score is posted, there is a one-week regrading period where mistakes can be corrected (or by the last day of class, whichever comes first). After the one-week timeframe, the posted score becomes final.

Notification of Recording

We will record the class sessions, or portions of them, for students who are unable to attend. Recording by other means is not permitted. The recorded class sessions will be posted on our Canvas class website.

If you have privacy concerns and do not wish to appear in the recording, please avoid sitting in the front section of the classroom, as the video camera may record that area of the room.

IV. COURSE SCHEDULE

For a detailed schedule with assignments, assessments, and due dates, see the Course Schedule tab, which is also linked from Canvas.

V. ACCOMMODATIONS

Disability Accommodation Procedures

UIC is committed to full inclusion and participation of people with disabilities in all aspects of university life. If you face or anticipate disability-related barriers while at UIC, please connect with the Disability Resource Center (DRC) at drc.uic.edu, via email at drc@uic.edu, or call (312) 413-2183 to create a plan for reasonable accommodations. To receive accommodations, you will need to disclose the disability to the DRC, complete an interactive registration process with the DRC, and provide us with a Letter of Accommodation (LOA). Upon receipt of an LOA, we will gladly work with you and the DRC to implement approved accommodations.

Religious Accommodations

Following campus policy, if you wish to observe religious holidays, you must notify us by the tenth day of the semester. If the religious holiday is observed on or before the tenth day of the semester, you must notify us at least five days before you will be absent. Please submit this form by email with the subject heading: "YOUR NAME: Requesting Religious Accommodation."

Extenuating Circumstances: "Don't Suffer in Silence"

We recognize that our students come from varied backgrounds and can have widely-varying circumstances. We also acknowledge that the circumstances of this particular semester may bring unique challenges. If you have any unforeseen circumstances that arise during the

course, please do not hesitate to contact the course staff or the instructor to discuss your situation. The sooner we are made aware, the more easily we can provide accommodations.

Typically, extenuating circumstances include work-school balance, familial responsibilities, health concerns, or anything else beyond your control that may negatively impact your performance in the class. Additionally, while some amount of "productive struggle" is healthy for learning, you should ask the course staff for help if you have been stuck on an issue for a very long time.

Life happens! While our focus is providing an excellent educational environment, our course does not exist in a vacuum. Our ultimate goal as a course staff is to provide you with the ability to be successful, and we encourage you to work with us to make that happen.

VI. CLASSROOM ENVIRONMENT

Inclusive Community

UIC values diversity and inclusion. Regardless of age, disability, ethnicity, race, gender, gender identity, sexual orientation, socioeconomic status, geographic background, religion, political ideology, language, or culture, we expect all members of this class to contribute to a respectful, welcoming, and inclusive environment for every other member of our class. We strive to create a challenging but inclusive environment that is conducive to learning for all students. If aspects of this course result in barriers to your inclusion, engagement, accurate assessment, or achievement, please notify us as soon as possible. If at any time you feel that you are not experiencing an inclusive environment, or you are made to feel uncomfortable, disrespected, or excluded, please report the incident so that we may address the issue and maintain a supportive and inclusive learning environment. You may contact the course staff or the CS academic advisors to express your concerns.

Name and Pronoun Use

If your name does not match the name on my class roster, please let the course staff know as soon as possible. My pronouns are he/him. We welcome your pronouns if you would like to share them with us. For more information about pronouns, see this page: https://www.mypronouns.org/what-and-why.

Community Agreement/Classroom Conduct Policy

- Be present by turning off cell phones and removing yourself from other distractions.
- Be respectful of the learning space and community. For example, no side conversations or unnecessary disruptions.
- Use preferred names and gender pronouns.
- Assume goodwill in all interactions, even in disagreement.

- Facilitate dialogue and value the free and safe exchange of ideas.
- Try not to make assumptions, have an open mind, seek to understand, and not judge.
- Approach discussion, challenges, and different perspectives as an opportunity to "think out loud," learn something new, and understand the concepts or experiences that guide other people's thinking.
- Debate the concepts, not the person.
- Be gracious and open to change when your ideas, arguments, or positions do not work or are proven wrong.
- Be willing to work together and share helpful study strategies.
- Be mindful of one another's privacy, and do not invite outsiders into our classroom.

VII. RESOURCES: Academic Success, Wellness, and Safety

We all need the help and the support of our UIC community. Please visit my drop-in hours for course consultation and other academic or research topics. For additional assistance, please contact your assigned college advisor and visit the support services available to all UIC students.

Academic Success

- UIC Tutoring Resources
- College of Engineering tutoring program
- Equity and Inclusion in Engineering Program
- UIC Library and UIC Library Research Guides.
- Offices supporting the UIC Undergraduate Experience and Academic Programs.
- Student Guide for Information Technology
- First-at-LAS Academic Success Program, focusing on LAS first-generation students.

Wellness

- **Counseling Services:** You may seek free and confidential services from the Counseling Center at https://counseling.uic.edu/.
- Access U&I Care Program for assistance with personal hardships.

- **Campus Advocacy Network:** Under Title IX, you have the right to an education that is free from any form of gender-based violence or discrimination. To make a report, email TitleIX@uic.edu. For more information or confidential victim services and advocacy, visit UIC's Campus Advocacy Network at http://can.uic.edu/.

Safety

- UIC Safe App — PLEASE DOWNLOAD FOR YOUR SAFETY!
- UIC Safety Tips and Resources
- Night Ride
- Emergency Communications: By dialing 5-5555 from a campus phone, you can summon the Police or Fire for any on-campus emergency. You may also set up the complete number, (312) 355-5555, on speed dial on your cell phone.